

A Directional-Derivative-Constrained Method for Continuously Steerable Differential Beamformers with Uniform Circular Arrays

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Abstract— This paper investigates the design of continuously steerable differential beamformers using uniform circular microphone arrays and proposes a method based on directional derivative constraints. By enforcing the first-order derivatives of the beampattern to be zero at the desired steering direction and appropriately specifying higher-order derivatives, the beamformer is guaranteed to achieve effective steering performance. This formulation not only enhances steering flexibility but also provides a more intuitive and robust framework for beampattern design. Simulation results confirm that the proposed method yields continuously steerable beampatterns.

Index Terms—Differential microphone arrays (DMAs), uniform circular arrays (UCAs), differential beamforming, derivative constraints, null steering.

I INTRODUCTION

Beamforming with microphone arrays plays a fundamental role in speech and acoustic signal processing applications such as teleconferencing, hearing aids, and human-machine interaction [1-5]. Among the various beamforming techniques proposed in the literature, differential beamforming has attracted considerable attention due to its compatibility with compact array structures and its inherently high directivity [6-10].

In general, differential beamformer design is carried out by enforcing unit gain in the desired direction while imposing nulls in undesired directions, either in the time domain or the short-time-Fourier-transform (STFT) domain [11-13]. However, linear differential microphone arrays (DMAs) typically exhibit limited steering capability and achieve optimal performance primarily in endfire directions [11], [14].

To enable full azimuthal steering, various array structures have been proposed, including linear superarrays composed of omnidirectional and bidirectional microphones [15], [16], circular DMAs [17-21], and other planar configurations [19], [20], [22], [23]. A common design principle for circular DMAs, inherited from linear DMAs, is to enforce unit gain in the target direction while placing nulls at undesired directions. However, this strategy often results in undesirable beampatterns and inaccurate steering performance [24].

Previous studies have attempted to address this issue by introducing symmetry constraints derived from the geometry of uniform circular arrays (UCAs) [17]. However, these constraints typically restrict beam steering to a limited set of predefined directions. Other methods, such as the series expansion approach [25], [26], provide full steering flexibility

but require prior knowledge of the complete desired directivity pattern, which is often unavailable in practice.

Another line of work extends null-constrained designs by exploiting the inherent symmetry of UCA configurations to achieve fully steerable beamformers [24], [27]. However, this approach lacks theoretical rigor and intuitive interpretability, as it relies primarily on enforcing symmetric null placement without a clear derivation or justification of the underlying symmetry constraints.

To overcome these limitations, this paper proposes a design method based on directional derivative constraints. By enforcing the first-order derivative of the beampattern to be zero at the desired steering direction and appropriately specifying higher-order derivatives, the proposed method establishes a stationary-point condition at the desired steering direction. Together with distortionless response and null constraints, this formulation guarantees a mainlobe peak in the desired steering direction and enables continuous beam steering. The resulting constraint provides a simple and intuitive mechanism for achieving accurately aligned and continuously steerable beams toward the desired source. Simulation results validate the theoretical analysis and demonstrate the effectiveness and robustness of the proposed method.

II SIGNAL MODEL AND PROBLEM FORMULATION

Consider a UCA, with radius r , consisting of M omnidirectional microphones. The center of the array is located at the origin of the Cartesian coordinate system, with azimuth angles measured counterclockwise from the x -axis. The position of the m th microphone is given by $\mathbf{r}_m = [r \cos \psi_m \ r \sin \psi_m]^T$, where $\psi_m = 2\pi(m-1)/M$, for $m = 1, 2, \dots, M$. With this configuration, the steering vector for the microphone array can be expressed as [28]

$$\mathbf{d}_\theta(\omega) = \begin{bmatrix} e^{j\varpi \cos(\theta - \psi_1)} & e^{j\varpi \cos(\theta - \psi_2)} & \dots & e^{j\varpi \cos(\theta - \psi_M)} \end{bmatrix}^T, \quad (1)$$

where j represents the imaginary unit and $\varpi = \omega r/c$, with $\omega = 2\pi f$ being the angular frequency, c the speed of sound in air, and $f > 0$ the temporal frequency.

Suppose a desired source, denoted as $X(f)$, propagates to the array from the angle θ_s . Neglecting relative multipath propagation and attenuation effects, the observed signal in the frequency domain can be represented as a vector of length

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M , given by

$$\begin{aligned} \mathbf{y}(\omega) &= [Y_1(\omega) \ Y_2(\omega) \ \cdots \ Y_M(\omega)]^T \\ &= \mathbf{d}_{\theta_s}(\omega) X(\omega) + \mathbf{v}(\omega), \end{aligned} \quad (2)$$

where $\mathbf{v}(\omega)$ is the zero-mean additive noise signal vector defined analogously to $\mathbf{y}(\omega)$. The beamforming process can be presented as

$$\begin{aligned} Z(\omega) &= \mathbf{h}^H(\omega) \mathbf{y}(\omega) \\ &= \mathbf{h}^H(\omega) \mathbf{d}_{\theta_s}(\omega) X(\omega) + \mathbf{h}^H(\omega) \mathbf{v}(\omega), \end{aligned} \quad (3)$$

where the superscript H denotes the conjugate transpose operator and $Z(\omega)$ is the estimate of the desired signal.

In the beamforming process, it is essential to apply the distortionless constraint to ensure that the desired signal from the steering direction passes through the beamformer without distortion. This constraint is implemented in this work as follows:

$$\mathbf{h}^H(\omega) \mathbf{d}_{\theta_s}(\omega) = 1. \quad (4)$$

To evaluate beamforming performance, we utilize three standard metrics commonly employed in the literature: beam-pattern, $\mathcal{B}[\mathbf{h}(\omega), \theta]$, white noise gain (WNG), $\mathcal{W}[\mathbf{h}(\omega)]$, and directivity factor (DF), $\mathcal{D}[\mathbf{h}(\omega)]$, which are expressed as follows [11]:

$$\mathcal{B}[\mathbf{h}(\omega), \theta] = \mathbf{h}^H(\omega) \mathbf{d}_{\theta}(\omega), \quad (5)$$

$$\mathcal{W}[\mathbf{h}(\omega)] = \frac{|\mathbf{h}^H(\omega) \mathbf{d}_{\theta_s}(\omega)|^2}{\mathbf{h}^H(\omega) \mathbf{h}(\omega)}, \quad (6)$$

and

$$\mathcal{D}[\mathbf{h}(\omega)] = \frac{|\mathbf{h}^H(\omega) \mathbf{d}_{\theta_s}(\omega)|^2}{\mathbf{h}^H(\omega) \mathbf{\Gamma}_d(\omega) \mathbf{h}(\omega)}, \quad (7)$$

where $\mathbf{\Gamma}_d(\omega)$ is the coherence matrix of the diffuse noise, whose (i, j) th element is $\text{sinc}(\omega \delta_{ij}/c)$, with $\delta_{ij} = 2r |\sin[(i-j)\pi/M]|$ being the distance between microphones i and j .

With the above formulation, the core challenge in beamforming is to determine the beamforming filter $\mathbf{h}$. Although various optimization techniques can be employed to identify the beamforming filter, in this work, we focus on designing differential beamformers such that the resulting beampattern closely matches the frequency-independent directivity pattern of an N th-order differential beamformer. This directivity pattern is given along θ_s by

$$\mathcal{B}_{N, \theta_s}(\theta) = \sum_{n=0}^N a_{N,n} \cos^n(\theta - \theta_s), \quad (8)$$

where $a_{N,n} \in \mathbb{R}$, for $n = 0, 1, \dots, N$, are coefficients satisfying $\sum_{n=0}^N a_{N,n} = 1$, which define the specific shape of the directivity pattern.

As seen from (8), the directivity pattern is symmetric with respect to θ_s , meaning that

$$\mathcal{B}_{N, \theta_s}(\theta_s + \theta') = \mathcal{B}_{N, \theta_s}(\theta_s - \theta'), \quad \theta' \in [0, \pi]. \quad (9)$$

Therefore, when designing a differential beamformer, the main beam should be directed towards θ_s , and the beampattern must exhibit symmetry about the line $\theta_s \leftrightarrow \theta_s + \pi$ for $\theta \in [0, \pi]$, i.e., $\mathcal{B}[\mathbf{h}(\omega), \theta_s + \theta] = \mathcal{B}[\mathbf{h}(\omega), \theta_s - \theta]$. By dividing the desired beampattern into two hemispheres centered on the look direction, if a null at $\theta_{N,n}^+$ exists in

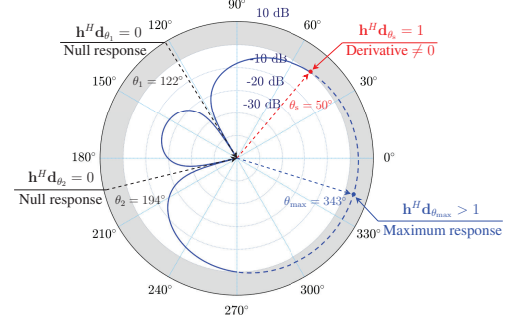


Fig. 1: Beampattern of a beamformer designed using the null-constrained method. The design parameters are as follows: a UCA with $M = 8$ microphones and a radius of $r = 2.0$ cm, operating at a frequency of $f = 1$ kHz, with a steering direction at $\theta_s = 50^\circ$, and two nulls located at 122° and 194° .

the upper hemisphere (i.e., within $[\theta_s, \theta_s + \pi]$), then, due to symmetry, a corresponding null must also appear at $\theta_{N,n}^-$ in the lower hemisphere (i.e., within $[\theta_s - \pi, \theta_s]$). This relationship satisfies: $(\theta_{N,n}^+ + \theta_{N,n}^-)/2 = \theta_s$. As a result, the following symmetry constraints should be enforced on the beampattern [24]:

$$\mathcal{B}[\mathbf{h}(\omega), \theta_{N,n}^+] = \mathbf{d}_{\theta_{N,n}^+}^H(\omega) \mathbf{h}(\omega) = 0, \quad (10)$$

$$\mathcal{B}[\mathbf{h}(\omega), \theta_{N,n}^-] = \mathbf{d}_{\theta_{N,n}^-}^H(\omega) \mathbf{h}(\omega) = 0. \quad (11)$$

By incorporating the null constraints, symmetry constraints, and the distortionless constraint, a linear system of equations can be derived and solved to obtain the beamformer [24]. While beam steering can be accomplished by applying symmetric null constraints, this approach lacks both theoretical clarity and intuitive interpretability. In contrast, this paper introduces a more effective design method, grounded in a thorough theoretical analysis, to provide clearer insights and improved performance.

III STEERABLE DIFFERENTIAL BEAMFORMERS VIA DIRECTIONAL DERIVATIVE CONSTRAINTS

Generally, an N th-order differential beamformer is expected to exhibit N nulls. Following the conventional design of linear DMAs, we begin by enforcing unit gain in the desired direction and zero response at the specified null directions.

To illustrate this, Fig. 1 shows examples using a UCA with eight sensors uniformly arranged on a circle of radius $r = 2.0$ cm. We aim to design a second-order hypercardioid beamformer with two initial nulls placed at 72° and 144° when the steering direction is $\theta_s = 0^\circ$. When the beamformer is steered to $\theta_s = 50^\circ$, the nulls are correspondingly shifted to 122° and 194° , where the null constraints are enforced. As shown, although the resulting beampatterns satisfy the imposed constraints, achieving unit gain in the desired direction (50°) and nulls at the specified locations (122° and 194°), they exhibit gains exceeding one in other directions. This behavior can lead to undesired amplification of noise, interference, or reverberation instead of suppression. The underlying issue stems from the inherent symmetry of the circular array, which is not properly accounted for in this beamforming design.

Upon closer inspection, we identify a key limitation in the above design: enforcing only unit response in the desired direction together with nulls at prescribed angles does not guarantee that the main lobe is centered at the steering direction. Although these constraints are satisfied, the maximum response may still deviate from the desired direction. Motivated by this observation, we adopt an alternative constraint strategy. Instead of constraining only the beampattern value at θ_s , we additionally enforce that its first-order derivative with respect to direction at θ_s is zero. This imposes a stationary-point condition at the desired direction and provides a more direct and reliable basis for accurate beam steering.

We first consider the design of a first-order differential beamformer with a single null at $\theta_{1,n}$. Under this setting, the desired directivity pattern in (8) simplifies to

$$\mathcal{B}_1(\theta) = a_{1,0} + a_{1,1} \cos(\theta - \theta_s). \quad (12)$$

The first-order derivative of $\mathcal{B}_1(\theta)$ with respect to θ is given by

$$\mathcal{B}_1^{(1)}(\theta) = a_{1,1} \sin(\theta - \theta_s). \quad (13)$$

One can check that $\mathcal{B}_1^{(1)}(\theta_s) = 0$. Consequently, the partial derivative of the beamformer's beampattern with respect to θ at θ_s should also satisfy

$$\left. \frac{\partial \mathcal{B}[\mathbf{h}(\omega), \theta]}{\partial \theta} \right|_{\theta=\theta_s} = 0, \quad (14)$$

which is equivalent to

$$[\mathbf{d}_{\theta_s}^H(\omega)]^{(1)} \mathbf{h}(\omega) = 0, \quad (15)$$

where $[\mathbf{d}_{\theta_s}^H(\omega)]^{(1)}$ denotes the first-order derivative of $\mathbf{d}_{\theta}^H(\omega)$ evaluated at θ_s . Under this condition, the null-constrained design of a first-order differential beamformer reduces to solving the following linear system of equations:

$$\mathbf{D}_C(\omega) \mathbf{h}(\omega) = \mathbf{i}_1, \quad (16)$$

where

$$\mathbf{D}_C(\omega) = \begin{bmatrix} \mathbf{d}_{\theta_s}^H(\omega) \\ [\mathbf{d}_{\theta_s}^H(\omega)]^{(1)} \\ \mathbf{d}_{\theta_{1,1}}^H(\omega) \end{bmatrix} \in \mathbb{C}^{3 \times M} \quad (17)$$

is the constraint matrix that incorporates the distortionless, null, and directional derivative constraints, and $\mathbf{i}_1 \in \mathbb{R}^{3 \times 1}$ is a unit vector whose first element is 1 and all remaining elements are 0.

To ensure clarity and avoid potential misinterpretation, we emphasize that enforcing a zero first-order directional derivative of the beampattern at the steering direction θ_s is a necessary but not sufficient condition for a local maximum. The local maximum at θ_s is ensured jointly by the distortionless constraint (which enforces unit gain) and the derivative constraints.

For the more general case of an N th-order differential beamformer, the beampattern is an N th-order function of θ . To ensure that the maximum response occurs at the steering direction θ_s , the first-order derivative of the beampattern with respect to θ must be zero at θ_s .

For higher-order (greater than 1) derivatives, they may either be computed analytically or specified based on design criteria. For instance, if a desired beampattern is available, the corresponding derivatives can be derived accordingly.

Alternatively, all odd-order derivatives can be constrained to 0 (which can be shown theoretically), while even-order derivatives (e.g., the second order) can be constrained to negative values, with their magnitudes influencing the beampattern shape, potentially leading to asymmetric patterns.

In this work, we focus on demonstrating that differential beamformers can be designed through derivative-based constraints, while the optimization of derivative values for precise beampattern shape control is left for future investigation. By combining the distortionless constraint, null constraints, and directional derivative constraints, the design reduces to solving the following linear system of equations:

$$\mathbf{D}_C(\omega) \mathbf{h}(\omega) = \mathbf{i}_\beta, \quad (18)$$

where

$$\mathbf{D}_C(\omega) = \begin{bmatrix} \mathbf{d}_{\theta_s}^H(\omega) \\ [\mathbf{d}_{\theta_s}^H(\omega)]^{(1)} \\ \vdots \\ [\mathbf{d}_{\theta_s}^H(\omega)]^{(N)} \\ \mathbf{d}_{\theta_{N,1}}^H(\omega) \\ \vdots \\ \mathbf{d}_{\theta_{N,N}}^H(\omega) \end{bmatrix} \in \mathbb{C}^{(2N+1) \times M} \quad (19)$$

is the constraint matrix with

$$[\mathbf{d}_{\theta_s}^H(\omega)]^{(q)} = \left. \frac{\partial^q \mathbf{d}_{\theta}^H(\omega)}{\partial \theta^q} \right|_{\theta=\theta_s}, \quad (20)$$

for $q = 1, 2, \dots, Q-1$, being the q th-order partial derivative of $\mathbf{d}_{\theta}^H(\omega)$ at θ_s , and $\mathbf{i}_\beta \in \mathbb{R}^{(2N+1) \times 1}$ is again a unit vector.

In some cases, a null may have multiplicity greater than one, allowing additional constraints to be imposed by enforcing that the derivatives at the null location are zero [29]. For instance, if a null at $\theta_{N,n}$ has multiplicity Q , the corresponding constraints can be expressed as

$$[\mathbf{d}_{\theta_{N,n}}^H(\omega)]^{(q)} \mathbf{h}(\omega) = 0, \quad (21)$$

for $q = 0, 1, \dots, Q-1$.

It can be verified that at least $2N+1$ microphones are required to design a steerable N th-order differential beamformer. This is consistent with the result in [25], which states that a minimum of $M = 2N+1$ microphones is necessary to achieve continuous steerability. A straightforward approach to obtaining the beamformer by solving (18) is to maximize the WNG, for which the solution is given by

$$\mathbf{h}_{\text{MWNG}}(\omega) = \mathbf{D}_C^H(\omega) [\mathbf{D}_C(\omega) \mathbf{D}_C^H(\omega)]^{-1} \mathbf{i}_\beta. \quad (22)$$

IV SIMULATIONS

In this section, we assess the performance of the proposed method for designing steerable differential beamformers (DMA-DerivCon) based on directional derivative constraints. We consider a UCA with a radius of $r = 2.0$ cm, comprising 8 omnidirectional microphones. For comparison, we also include the conventional null-constrained method (DMA-Null) [11], the symmetric null-constrained method (DMA-SymNull) [24], and the series expansion method (DMA-SeriesExp) [25].

In the first simulation, we investigate the design of a first-order differential beamformer. The steering direction is set

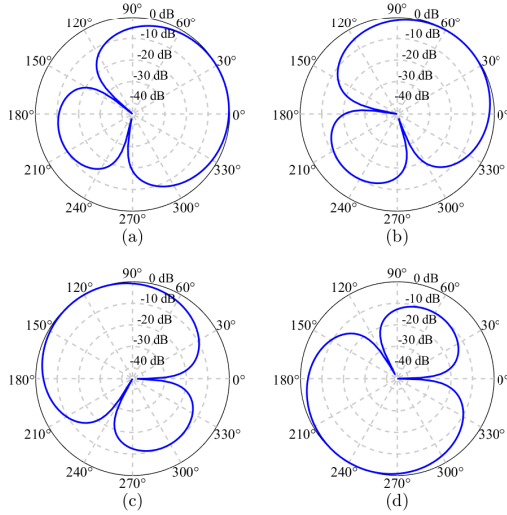


Fig. 2: Beampatterns at $f = 1$ kHz for the first-order differential beamformer designed using the proposed method: (a) $\theta_s = 20^\circ$, (b) $\theta_s = 50^\circ$, (c) $\theta_s = 120^\circ$, and (d) $\theta_s = 240^\circ$.

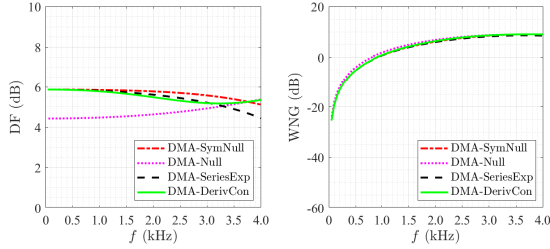


Fig. 3: DF and WNG of first-order differential beamformers designed using DMA-Null, DMA-SymNull, DMA-SeriesExp, and DMA-DerivCon. Conditions are $M = 8$, UCA with $r = 2.0$ cm, and $\theta_s = 50^\circ$.

to $\theta_s = 20^\circ, 50^\circ, 120^\circ$, and 240° , and the corresponding null is placed at $\theta_s + 120^\circ$, with $\mathbf{i}_\beta = [1 \ 0 \ 0]^T$. Figure 2 shows the resulting beampatterns. It can be observed that the main lobes are correctly steered toward the desired directions, while deep nulls are formed at the expected locations.

Figure 3 compares the DF and WNG of the four methods under consideration, including DMA-Null, DMA-SymNull, DMA-SeriesExp, and the proposed DMA-DerivCon. As shown, the proposed DMA-DerivCon achieves a more balanced DF across the entire frequency band while maintaining a WNG comparable to that of the other methods.

In the second simulation, we investigate the design of a second-order differential beamformer. The steering direction is again set to $\theta_s = 20^\circ$, while the nulls are placed at $\theta_s + 120^\circ$ and $\theta_s + 240^\circ$, respectively, with $\mathbf{i}_\beta = [1 \ 0 \ -2 \ 0 \ 0]^T$. Figure 4 plots the resulting beampatterns. As in the previous simulation, it can be observed that the main lobes are accurately steered toward the desired directions, while deep nulls are formed at the expected interference angles. This validates the ability of the proposed method to impose multiple directional constraints while maintaining a distortionless response in the target direction.

Figure 5 compares the DF and WNG of the four methods under consideration. As shown, the proposed DMA-DerivCon maintains a relatively high and smooth DF across the frequency band while achieving a WNG comparable to that

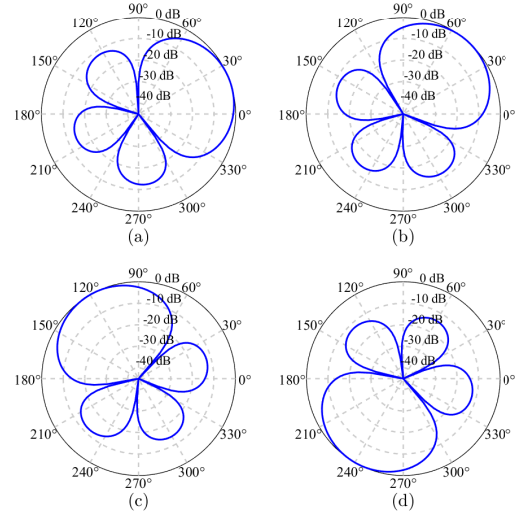


Fig. 4: Beampatterns at $f = 1$ kHz for the second-order differential beamformer designed using the proposed method: (a) $\theta_s = 20^\circ$, (b) $\theta_s = 50^\circ$, (c) $\theta_s = 120^\circ$, and (d) $\theta_s = 240^\circ$.

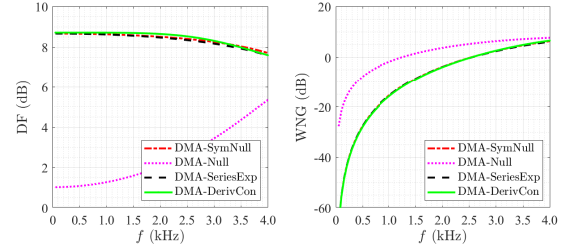


Fig. 5: DF and WNG of second-order differential beamformers designed using DMA-Null, DMA-SymNull, DMA-SeriesExp, and DMA-DerivCon. Conditions are $M = 8$, UCA with $r = 2.0$ cm, and $\theta_s = 50^\circ$.

of the other methods. This demonstrates that the derivative-constrained design remains effective for higher-order differential beamforming without noticeable degradation in noise robustness.

V CONCLUSIONS

Designing steerable beamformers with UCAs is crucial for many sound capture applications. In this paper, a directional derivative-constrained method was proposed for the design of steerable differential beamformers with UCAs. It was demonstrated that constraining only the desired and null directions failed to guarantee correct steering, as the maximum response might not appear in the intended direction. To address this issue, the proposed method introduced directional derivative constraints by forcing the first-order derivatives of the beampattern at the desired steering direction to zero and assigning suitable values to higher-order derivatives. This ensured that the beamformer achieved its maximum response in the target direction and provided continuous steering capability. Simulation results validated the theoretical analysis and confirmed that the proposed beamformer could produce well-formed beampatterns with accurate steering. The method, therefore, offers a practical and effective solution for steerable differential beamforming in circular array configurations.

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